
Formatting Instructions For NeurIPS 2026

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Affiliation

Address

email

Abstract

The abstract paragraph should be indented 1/2 inch (3 picas) on both the left- and right-hand margins. Use 10 point type, with a vertical spacing (leading) of 11 points. The word **Abstract** must be centered, bold, and in point size 12. Two line spaces precede the abstract. The abstract must be limited to one paragraph.

1 Introduction

Word-of-mouth and viral marketing techniques have become dominant strategies for product promotion in social networks. By selecting a small set of influential seed users, companies can trigger cascades of adoption along social connections, a problem commonly referred to as influence maximization (IM). Over the past two decades, IM has been extensively studied under classical models such as the Independent Cascade (IC) and Linear Threshold (LT) models, where activated nodes attempt to influence their neighbors in a broadcast manner. However, many real-world campaigns deviate from this paradigm. For example, when a retailer distributes digital coupons, a user who chooses not to redeem or discard the coupon may forward it to at most one other user. A similar pattern arises in invitation-code campaigns, where an inactive user can pass the code to only one specific friend. In both scenarios, the promotional content propagates via a sequential, one-to-one diffusion process. We refer to this setting as sequential influence diffusion (SID), which admits a formulation closely related to random walks on the social graph. Despite its practical relevance, this setting remains underexplored, with existing IM frameworks inherently tailored to broadcast diffusion and unable to capture such sequential propagation mechanisms.

This sequential propagation fundamentally alters both the diffusion mechanism and the objective landscape. First, the diffusion model is structurally different. In standard IC and LT models, an activated node attempts to influence all its neighbors simultaneously, resulting in a branching cascade that fans out rapidly across the network. In SID, however, each piece of promotional content is passed to at most one neighbor, tracing a chain-like propagation process through the network. Second, the definition of activation is strictly decoupled from mere exposure. In classical IM, a node is considered activated as soon as the message reaches it. In coupon or invitation-code scenarios, however, a node that simply forwards the coupon without using it is not considered activated; only a node that ultimately redeems the coupon counts as an activated user. Consequently, the intermediate nodes along the forwarding chain contribute nothing to the final activation. Third, since multiple coupons propagate independently, the same user may redeem more than one coupon, yet is counted as activated only once. The primary goal is therefore to maximize the number of distinct users who ultimately redeem at least one coupon, as each newly activated user represents a potential recurring customer. Beyond this, when multiple seeding strategies yield the same activation coverage, the total number of coupons redeemed also serves as an additional metric. These differences collectively fall outside the modeling scope of existing IM frameworks and call for a new problem formulation.

To address these challenges, we formalize the *Sequential Influence Diffusion Influence Maximization (SIDIM)* problem. In our model, k coupons are distributed to k seed nodes, and each coupon propagates independently through the network as a random walk. At each node along the walk, one

of three outcomes occurs: the node redeems the coupon and becomes activated, the node discards the coupon, or the node forwards the coupon to exactly one randomly selected neighbor. In the first two cases, the random walk terminates; in the third, it continues. The objective is to select k seed nodes that maximize the expected number of distinct users who redeem at least one coupon. Since the k coupons propagate independently, each coupon induces its own possible world along the nodes it may visit. The overall diffusion outcome is then determined by the joint configuration of k independent possible worlds. We first prove that SIDIM is NP-hard, confirming the inherent computational intractability of the problem. We then analyze the structural properties of the objective function through the possible-world framework and establish that the expected adoption spread is monotone and submodular. This key property enables the application of a greedy algorithm with a $(1 - 1/e)$ -approximation guarantee. To make the greedy algorithm scalable, we develop a novel Reverse Influence Sampling (RIS) framework tailored to the sequential diffusion setting. Unlike classical RR set construction, where a single RR set captures all potential influence paths to a target node, our framework jointly samples k reverse reachable sets for each target — one per coupon, corresponding to one per possible world — reflecting the independent random walks in the diffusion process. Building on this RIS framework, we design a near-linear time seed selection algorithm with a $(1 - 1/e - \epsilon)$ -approximation guarantee.

Our contributions are summarized as follows. First, we initiate the formal study of the SIDIM problem, a practically motivated yet theoretically underexplored setting that captures the single-path, multi-coupon nature of real-world sequential promotional campaigns. Second, we prove that SIDIM is NP-hard and establish that the adoption spread function is monotone and submodular, admitting a greedy algorithm with a $(1 - 1/e)$ -approximation guarantee. Third, we develop a tailored RIS scheme that jointly generates k reverse reachable sets per root node with provable unbiasedness, and build upon it a near-linear time algorithm with a $(1 - 1/e - \epsilon)$ -approximation guarantee. Fourth, extensive experiments on real-world social networks demonstrate that our method significantly outperforms baselines adapted from classical IM, validating both the importance of the sequential diffusion formulation and the effectiveness of our algorithmic framework.

2 Submission of papers to NeurIPS 2026

Please read the instructions below carefully and follow them faithfully.

2.1 Style

Papers to be submitted to NeurIPS 2026 must be prepared according to the instructions presented here. Papers may only be up to **nine** pages long, including figures. **Papers that exceed the page limit will not be reviewed (or in any other way considered) for presentation at the conference.** Additional pages *containing acknowledgments, references, checklist, and optional technical appendices* do not count as content pages.

The margins in 2026 are the same as those in previous years.

Authors are required to use the NeurIPS L^AT_EX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for desk rejection.

2.2 Retrieval of style files

The style files for NeurIPS and other conference information are available on the website at

<https://neurips.cc>.

The only supported style file for NeurIPS 2026 is `neurips_2026.sty`, rewritten for L^AT_EX 2_ε. **Previous style files for L^AT_EX 2.09, Microsoft Word, and RTF are no longer supported.**

The L^AT_EX style file contains three optional arguments:

- `final`, which creates a camera-ready copy,
- `preprint`, which creates a preprint for submission to, e.g., arXiv,
- `nonatbib`, which will not load the `natbib` package for you in case of package clash.

87 **Preprint option** If you wish to post a preprint of your work online, e.g., on arXiv, using the
 88 NeurIPS style, please use the preprint option. This will create a nonanonymized version of your
 89 work with the text “Preprint. Work in progress.” in the footer. This version may be distributed as you
 90 see fit, as long as you do not say which conference it was submitted to. Please **do not** use the final
 91 option, which should **only** be used for papers accepted to NeurIPS.

92 At submission time, please omit the final and preprint options. This will anonymize your
 93 submission and add line numbers to aid review. Please do *not* refer to these line numbers in your
 94 paper as they will be removed during generation of camera-ready copies.

95 The file `neurips_2026.tex` may be used as a “shell” for writing your paper. All you have to do is
 96 replace the author, title, abstract, and text of the paper with your own.

97 The formatting instructions contained in these style files are summarized in Sections 4, 5, and 6
 98 below.

99 3 Problem Definition

100 Let $G = (V, E)$ be a directed graph with $n = |V|$ nodes and $m = |E|$ edges, where a node
 101 $u \in V$ represents a user and a directed edge $(u, v) \in E$ indicates that u may forward a coupon to v .
 102 Throughout this paper, we assume G is strongly connected. Given a seed set $S = \{s_1, s_2, \dots, s_k\} \subseteq$
 103 V of size k , a retailer distributes k coupons, one to each seed user, and each coupon then propagates
 104 independently through the network.

105 We first describe the propagation of a single coupon. When a coupon arrives at a node u , one of three
 106 mutually exclusive outcomes occurs. With probability p_u^a , user u *adopts* the coupon: the coupon is
 107 redeemed, u is marked as activated, and the propagation terminates at u . With probability p_u^d , user u
 108 *discards* the coupon: the coupon is removed from the network without activating anyone. With the
 109 remaining probability $p_u^t = 1 - p_u^a - p_u^d$, user u *transfers* the coupon to exactly one out-neighbor v ,
 110 selected according to an edge-level forwarding parameter $p(u, v)$ satisfying $\sum_{v:(u,v) \in E} p(u, v) = p_u^t$.
 111 A crucial feature of this process, which sharply distinguishes our setting from classical IM models,
 112 is that the coupon itself is *non-replicable*: unlike a piece of information in the IC or LT model that
 113 can be copied and broadcast to all neighbors simultaneously, a coupon exists as a single instance
 114 in the network and moves to at most one location at each step. When k coupons are distributed to
 115 the k seeds, they propagate independently according to the rule above, and a user u is considered
 116 ultimately *activated* if at least one of the k coupons is adopted at u .

117 *Remark 1.* In our model, the three outcomes at a node and the selection of the forwarding neighbor
 118 are not drawn from independent random variables; rather, they are jointly determined by a single
 119 uniform random number $X_u \in [0, 1]$ that partitions the unit interval into adoption, discarding, and
 120 per-neighbor transfer segments of widths p_u^a , p_u^d , and $\{p(u, v)\}_{v:(u,v) \in E}$, respectively. This contrasts
 121 with classical IM models, where the activation outcome on each edge is typically governed by its
 122 own independent coin flip.

123 Let $\mu(u, S) \in \{0, 1\}$ denote whether user u is activated under the seed set S , and let $I_C(S) =$
 124 $\mathbb{E}[\sum_{u \in V} \mu(u, S)]$ denote the expected number of activated users, taken over the randomness of the
 125 propagation process C . We refer to $I_C(S)$ as the *adoption spread* of S . The problem in this paper is
 126 formalized as:

127 **Problem Definition 1(Coupon Influence Maximization).** Given a directed graph $G = (V, E)$ with
 128 node parameters $\{(p_u^a, p_u^d)\}_{u \in V}$, edge forwarding parameters $\{p(u, v)\}_{(u,v) \in E}$, and an integer k ,
 129 find a size- k seed set S^* that maximizes the expected adoption spread, i.e.,

$$S^* = \arg \max_{S \subseteq V, |S|=k} I_C(S). \quad (1)$$

130 4 General formatting instructions

131 The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long.
 132 The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points.
 133 Times New Roman is the preferred typeface throughout, and will be selected for you by default.
 134 Paragraphs are separated by $\frac{1}{2}$ line space (5.5 points), with no indentation.

135 The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal
136 rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow $\frac{1}{4}$ inch
137 space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the
138 page.

139 For the final version, authors' names are set in boldface, and each name is centered above the
140 corresponding address. The lead author's name is to be listed first (left-most), and the co-authors'
141 names (if different address) are set to follow. If there is only one co-author, list both author and
142 co-author side by side.

143 Please pay special attention to the instructions in Section 6 regarding figures, tables, acknowledgments,
144 and references.

145 **5 Headings: first level**

146 All headings should be lower case (except for first word and proper nouns), flush left, and bold.

147 First-level headings should be in 12-point type.

148 **5.1 Headings: second level**

149 Second-level headings should be in 10-point type.

150 **5.1.1 Headings: third level**

151 Third-level headings should be in 10-point type.

152 **Paragraphs** There is also a `\paragraph` command available, which sets the heading in bold, flush
153 left, and inline with the text, with the heading followed by 1 em of space.

154 **6 Citations, figures, tables, references**

155 These instructions apply to everyone.

156 **6.1 Citations within the text**

157 The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as
158 long as you maintain internal consistency. As to the format of the references themselves, any style is
159 acceptable as long as it is used consistently.

160 The documentation for `natbib` may be found at

161 `http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

162 Of note is the command `\citet`, which produces citations appropriate for use in inline text. For
163 example,

164 `\citet{hasselmo}` investigated\dotso

165 produces

166 Hasselmo, et al. (1995) investigated...

167 If you wish to load the `natbib` package with options, you may add the following before loading the
168 `neurips_2026` package:

169 `\PassOptionsToPackage{options}{natbib}`

170 If `natbib` clashes with another package you load, you can add the optional argument `nonatbib`
171 when loading the style file:

172 `\usepackage[nonatbib]{neurips_2026}`

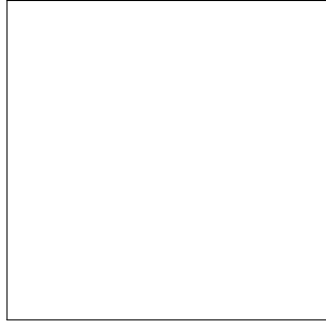


Figure 1: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

As submission is double blind, refer to your own published work in the third person. That is, use “In the previous work of Jones et al. [4],” not “In our previous work [4].” If you cite your other papers that are not widely available (e.g., a journal paper under review), use anonymous author names in the citation, e.g., an author of the form “A. Anonymous” and include a copy of the anonymized paper in the supplementary material.

6.2 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number¹ in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.²

6.3 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for the first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

6.4 Tables

All tables must be centered, neat, clean, and legible. The table number and title always appear before the table. See Table 1.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for the first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the booktabs package, which allows for typesetting high-quality, professional tables:

<https://www.ctan.org/pkg/booktabs>

This package was used to typeset Table 1.

6.5 Math

Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You

¹Sample of the first footnote.

²As in this example.

Table 1: Sample table caption. Explain what the table shows and add a key take-away message to the caption.

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	≈ 100
Axon	Output terminal	≈ 10
Soma	Cell body	up to 10^6

really shouldn't be using \$\$ anyway; see <https://tex.stackexchange.com/questions/503/why-is-preferable-to> and <https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath> for more information.)

6.6 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes. Please note that pages should be numbered.

7 Preparing PDF files

Please prepare submission files with paper size "US Letter," and not, for example, "A4."

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF file uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdf fonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

```
\usepackage{amsfonts}
```

followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

```
\newcommand{\RR}{I\!\!R} %real numbers
\newcommand{\Nat}{I\!\!N} %natural numbers
\newcommand{\CC}{I\!\!C} %complex numbers
```

Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

7.1 Margins in L^AT_EX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (<http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf>)

238 A number of width problems arise when L^AT_EX cannot properly hyphenate a line. Please give LaTeX
239 hyphenation hints using the \- command when necessary.

240 **References**

241 References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level
242 heading for the references. Any choice of citation style is acceptable as long as you are consistent. It
243 is permissible to reduce the font size to small (9 point) when listing the references. Note that the
244 Reference section does not count towards the page limit.

245 [1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In
246 G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp.
247 609–616. Cambridge, MA: MIT Press.

248 [2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the*
249 *GENeral NEural Simulation System*. New York: TELOS/Springer-Verlag.

250 [3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent
251 synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

252 **A Technical appendices and supplementary material**

253 Technical appendices with additional results, figures, graphs, and proofs may be submitted with the
254 paper submission before the full submission deadline (see above). You can upload a ZIP file for
255 videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the
256 technical appendices.

257 Note: Think of the appendix as “optional reading” for reviewers. The paper must be able to stand
258 alone without the appendix; for example, adding critical experiments that support the main claims to
259 an appendix is inappropriate.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”.
- Keep the checklist subsection headings, questions/answers and guidelines below.
- Do not modify the questions and only use the provided macros for your answers.

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The authors should answer **[Yes]** if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[TODO]**

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